

Executive Summary - Exploring Meaning Beyond the Axes: A Comparative Study of PCA Interpretability in High-Dimensional Representation Spaces

Tudor-Octavian Mihăiță

Understanding high-dimensional data has become a core challenge in modern machine learning and data analysis. Many representations, such as biological measurements, behavioral metrics, and learned embeddings, contain hundreds of dimensions that cannot be directly interpreted by humans. **Principal Component Analysis (PCA)** is one of the most widely used dimensionality-reduction tools for exploring such data [1], yet **its interpretability** varies, in some domains it reveals meaningful structure, while in others it reduces data to abstract combinations with no semantic value. This study investigates why this discrepancy occurs by examining how PCA behaves across two fundamentally different forms of high-dimensional representations and how effectively it captures patterns aligned with human-defined concepts.

Prior research shows that word embeddings encode semantic relations as linear vector directions [2], including analogy patterns and gender subspaces [4,5]. Such findings suggest that PCA may reveal meaningful concepts when applied to representations with strong linear structure. By contrast, tabular datasets composed of human-engineered variables typically lack such structure, making PCA components harder to interpret. This motivates the central **research question**: *How does PCA influence interpretability across different forms of high-dimensional data, and what types of latent information can it successfully recover?*

To explore this question, PCA was applied to two contrasting datasets. The first experiment used **300-dimensional GloVe word embeddings** [3], combined with definitional gender word pairs extracted from gender-classified lists of words to construct a concept-definitional matrix. PCA revealed a dominant semantic direction representing *gender*, explaining **75.7%** of the variance across pairs. When projecting male and female associated words onto this direction, a clear linear separation emerged, and a classifier using only PC1 achieved an AUC of **95.7%**, comparable to a logistic regression model trained on the full embedding space. These results demonstrate that PCA can function as a concept-discovery tool in learned semantic spaces where meaning is distributed linearly.

The second experiment applied PCA to the **Wine Quality Prediction dataset** [6], consisting of 11 physiochemical features. In this structured feature space, PCA behaved differently. The variance was spread across many components, with PC1 accounting for only **28.7%**. Two-dimensional projections showed no clear clustering when samples were colored by quality, and component loadings combined chemical features in ways that lacked domain coherence. Correlation analysis further confirmed that no PC aligned strongly with the quality label. In this context, PCA reduced dimensionality but did not capture interpretable latent factors.

Overall, the results demonstrate that *PCA is not inherently interpretable*. Instead, interpretability arises only when the underlying data contains strong, aligned linear structure. Learned word embeddings satisfy this condition, enabling PCA to recover meaningful semantic axes, while heterogeneous tabular datasets do not. This clarifies why PCA succeeds in some representation spaces but fails in others.

Future work may explore nonlinear dimensionality-reduction techniques such as t-SNE or UMAP, linear variants designed for interpretability such as Independent Component Analysis (ICA) or Sparse PCA, and broader cross-domain comparisons to more precisely map the boundary between structures PCA can uncover and those for which alternative techniques are required.

References

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